Masters Project Semester Report

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***Abstract*—This project develops and deploys a Rover Robotics platform for autonomous poultry-farm operations, with the goal of serving multiple research projects on chicken farm monitoring. For example, measuring chicken's travel distance based on the movement of the robot, or scooping up dead chickens. Modern poultry houses present a dense, dynamic, and sensitive environment where manual data collection or experiments are labor-intensive and inefficient. To address this , I am working on a mobile Rover Robot with LiDar-driven perception, ROS2 navigation, and a variety of configurable modules to support a range of operational tasks. Beyond locomotion and sensing, my central research objective is replicating and extending the Collaborative Direction of Arrival (CDOA)–based RSSI localization method proposed by Latif & Parasuraman [1], evaluating its practicality for real-world field robotics. The rover employs LiDAR perception, ROS2 navigation, and a configurable sensing stack while incorporating an RSSI-only localization module designed to operate in GPS-denied agricultural environments. The CDOA algorithm leverages cooperative RSSI measurements from distributed wireless nodes to estimate the robot's direction of arrival and refine its position using Bayesian inference (EM and PF).To date, testing has been conducted in the lab environments and an indoor testbed designed to approximate key features of a poultry house. My future work will involve full deployment and validation inside an operational poultry farm to assess robustness, animal interaction dynamics, and real-world research utility. This ongoing effort aims to establish an extensible robotic framework for precision indoor poultry navigation.**

***Index Terms*—Rover Robotics, localization, navigation, poultry**

I. Introduction

The poultry industry relies on continuous monitoring of bird health, behavior, and environmental conditions to maintain welfare standards and operational efficiency. However, modern commercial poultry houses present substantial challenges for manual observation. Large flock sizes, uneven terrains, and large scale environment make traditional data collection methods labor-intensive and is extremely difficult. As a result, researchers continue to seek automated methods, such as integrating robots that can support measurement of flock activity, to reduce the amount of work needed to be done manually.

Indoor localization is a foundational capability for mobile robots operating in GPS-denied environments such as commercial poultry houses. These environments are cluttered and noisy where LiDAR and camera-based localization can degrade, and where full SLAM pipelines may fail due to occlusions or rapidly changing scene geometry. For agricultural researchers, these limitations constrain the use of mobile robots for tasks such as monitoring flock activity, estimating chicken travel distance, and conducting automated welfare inspections.

Recent work in wireless-signal–based localization offers a promising alternative. Latif and Parasuraman's CDOA framework demonstrated that Received Signal Strength Indicator (RSSI) measurements from a small number of static wireless sensor nodes can be used to estimate the Direction of Arrival (DOA) of a mobile node, and that combining CDOA with Bayesian methods such as Expectation Maximization (EM) or Particle Filtering (PF) enables accurate indoor localization without fingerprinting or expensive sensors [1]. This approach is particularly efficient for agricultural robotics, where simplicity, low cost, and robustness to visual noise are critical.

My project aims to adapt, replicate, and scale the CDOA methodology for a Rover Robotics platform intended for poultry farm operations. The rover is designed as a multipurpose research tool capable of estimating flock mobility by using its own movement as a surrogate for bird travel distance, performing mortality retrieval tasks through mechanical attachments, and supporting modular environmental sensing. To enable reliable navigation and data collection, I am implementing a wireless localization system based on collaborative RSSI sampling across distributed access points. My goal is to assess whether the CDOA method can perform effectively in the challenging conditions of a real poultry house.

To date, experiments have been conducted in the lab settings and in an indoor testbed. These evaluations have focused on verifying the reproducibility of the CDOA algorithm and integrating the localization output with the rover's ROS2 navigation stack. Future work includes deploying the system inside an active poultry farm to examine localization stability under dynamic flock movement. Ultimately, this project aims to bridge wireless signal–based localization research with real-world agricultural robotics, establishing a pathway for low-cost, scalable robot navigation in commercial poultry production.

II. Literature Review

Indoor localization in GPS-denied environments remains a core challenge in mobile robotics, particularly in settings where visual sensors and LiDAR can fail due to rapid change in the environment geometry due to chicken's movements. Latif & Parasuraman (2023) [1] present one of the most

promising recent advances in wireless-signal–based localization: the Collaborative Direction of Arrival (CDOA) method, which enables instantaneous localization of mobile robotic nodes using only RSSI data from a small number of static wireless sensors (4 count).

The key contribution of their work is calculating the RSSI gradients, which are measured collaboratively across multiple wireless sensor nodes. This can be used to compute the estimate of the Direction of Arrival without requiring fingerprinting. This is achieved through the central finite-difference method, which uses spatially distributed RSSI readings to approximate the signal gradient along the x and y axes. The resulting gradient is then converted into a CDOA angle, providing a bearing estimate for the mobile robot.

A relevant aspect of this work is its focus on resource-constrained environment and low-cost wireless infrastructure. Unlike vision-based or LiDAR-based localization methods, CDOA does not rely on lighting or GPS. This method requires only simple wireless nodes and RSSI measurements, making it very efficient and robust in difficult environments.

Overall, the CDOA work by Latif and Parasuraman forms a strong foundation for robotics research seeking low-cost, scalable indoor localization without reliance on vision or LiDAR. Their findings suggest that wireless-only localization can outperform classical RSSI techniques and provide practical accuracy for mobile robot navigation. This is perfect for applications such as poultry-farm robotics, where the dynamic factors degrade visual and GPS-based methods.

III. Design and Implementation

The primary objective of this project is to replicated and extend the Collaborative Direction of Arrival (CDOA) indoor localization method proposed by Latif & Parasuraman [1] and integate it into a Rover Robot. The following subsections describe the system architecture, hardware setup, and software implementation. A total of nine wireless nodes are deployed to enable large and accurate coverage for RSSI measurement.

A. System Architecture Overview

The system consists of three main components:

- A mobile robot platform (Rover Robotics) equipped with:
  - A single wireless node operating as Access Point (AP)
  - ROS2 Jazzy
  - Teleoperation control
  - Data logging for odometry, RSSI, and localization output
- A wireless sensor network composed of eight Raspberry Pi 5 static wireless nodes, distributed throughout the indoor chicken farm
- A CDOA localization pipeline implemented in ROS2, which:
  - Receives RSSI measurements from all static nodes
  - Compute RSSI gradients
  - Calculate the CDOA angle
  - Perform localization using EM or Particle Filter
  - Publish estimated pose for downstream use (analysis or autonomous navigation)

This architecture mirrors the reference paper while scaling up the number of wireless nodes from four to nine to increase the spatial coverage for larger environments.

B. Visualization

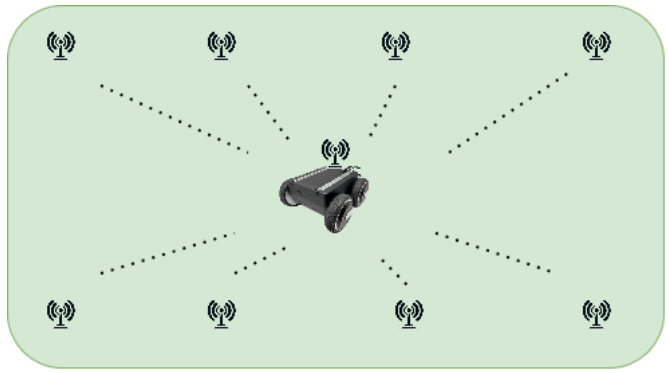

Fig. 1: Setup 1

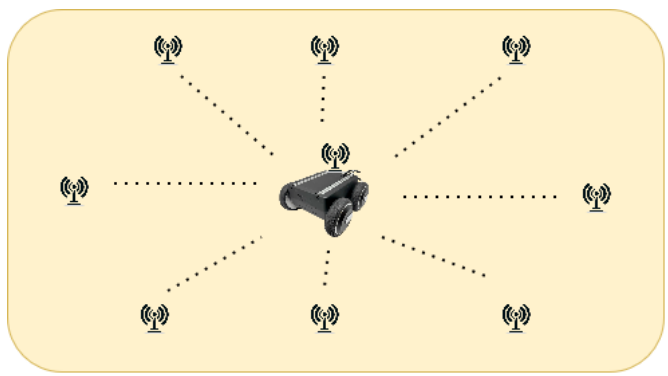

Fig. 2: Setup 2

The experimental setup consists of a teleoperated mobile robot positioned at the center of a distributed wireless sensor network composed of eight static nodes arranged around the workspace. Each static node measures the Wi-Fi RSSI broadcast from the access point to enable signal sampling for CDOA-based gradient estimation. The two configurations illustrated in Figures 1 and 2 represent variations in node spacing and geometric distribution designed to evaluate the sensitivity of the CDOA method to environmental layout. These controlled configurations provide a structured system for analyzing the impact of node placement on overall system performance within a ROS2 Jazzy framework.

C. Code Implementation

To support the CDOA localization algorithm, modifications were required to the existing `ros2-network-analysis` GitHub repository. The original package provides basic network diagnostics such as RSSI, link quality, packet errors,

and throughput; however, a lot of the code were not compatible with the new ROS2 Jazzy package. Therefore, several components of the repository were modified, specifically the `network_errors.py` file, which included deprecated syntax.

```
# segment errors at (tcp) protocol level
f = Popen(cmd1, shell=True, stdout=PIPE, stderr=PIPE)
msg.retransmits = int(f.stdout.read())  # TCP/IP segments retransmited
#node.get_logger.info("Total TCP Segments retransmited %d", msg.retransmits)
```

Fig. 3: Original `network_errors.py` File

Originally, this file contains an errors on line such as `msg.retransmits`. The problem was that this line is read from standard output and casted on an integer. However, the data type of read() is **byte**. Therfore, I have made changes throughout the codes to first decode it then cast after. Figure 4. shows the modified code.

```
# segment errors at (tcp) protocol level
f = Popen(cmd1, shell=True, stdout=PIPE, stderr=PIPE)
#print(f.stdout.read().decode().strip())
msg.retransmits = int(f.stdout.read().decode().strip())  # TCP/IP segments retransmited
node.get_logger().info(f"Total TCP Segments retransmited {msg.retransmits}")
```

Fig. 4: Modified `network_errors.py` File

Another problem with the localization was that the area size and the node boundary were set incorrectly. In the Figure 5., the area size and node positions were measured in meters. I have realized that I need to modify these to match the real-world measurements; this original code was used on a test-bed.

```
areaSize=(2.34, 1.75)

node_pos=[(0,0),(0,1.75),(2.34,1.75),(2.34,0)]
```

Fig. 5: Original `PFLocalization_tabletop.py` File

IV. Status Update

A. August: IMU Debugging

During August, the primary focus was resolving reliability issues with the BNO055 inertial measurement unit. When I started the Rover Robot driver, it kept on crashing and returned an error code. The core problem was that the BNO055 driver was both corrupted and the robot was reading the wrong ttyUSB port. I fixed it by redownloading the whole BNO055 driver and made sure that the correct port is read and permission is given.

B. September: Code Replication and Migration

In September, my work shifted to migrating the CDOA localization system from the original ROS implementation to ROS2 Jazzy. This involved replicating the existing algorithm structure, converting Python and C++ nodes to ROS2 communication primitives, replacing deprecated codes.The migration also required modifying launch files, restructuring build dependencies, and ensuring compatibility with colcon rather than catkin. By the end of the month, I have updated the repository with a functional ROS2-based version of the core algorithm. This is be very useful for other research projects too.

C. October: Wireless Node Setup

In October, I focused on constructing the wireless measurement infrastructure required for the CDOA method. A set of Raspberry Pi 5 devices was configured as static wireless nodes capable of receiving Wi-Fi signals from the robot-mounted access point. I installed Ubuntu 22.04 LTS, with ROS2 Jazzy on the nodes. Each Pi was set up with a lightweight ROS2 client to publish RSSI measurements in real time. This month also involved network configuration and testing across nodes. The finalized layout established the foundation for multi-node RSSI acquisition and gradient computation.

D. November: Full-System Integration Attempts

In November, all system components were integrated: the access point, static Pi nodes, ROS2 messaging system, CDOA gradient computation, and Bayesian localization modules. End-to-end algorithm runs were attempted on the actual rover robot. Although the pipeline executed, the localization output was unsuccessful and did not converge to the robot's true position. The problem might be either from the incorrect environment configuration or the algorithms. In Figure 3, it looks like the predicted trajectory overlapped with the actual odometry data.

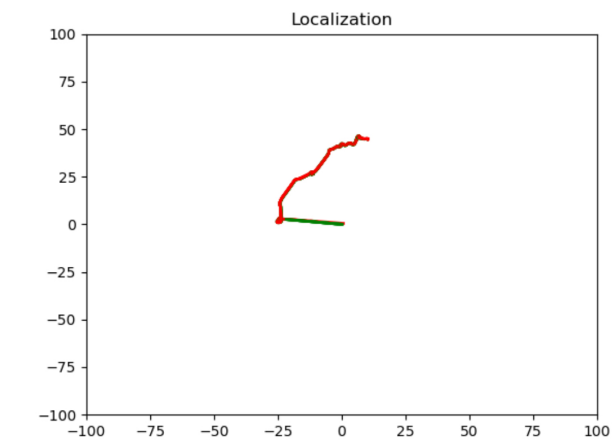

Fig. 6: Localization Output

E. December: Small-Scale Experiment

This month, I am focusing on replicating the full wireless localization setup on a smaller mobile robot platform (Husarion Rosbot), which enables for a more controlled experiment to assess the accuracy of the algorithm. The lab's testbed makes it easier to debug the system because of its small scale.

V. Discussion

The progress achieved throughout this project highlights both the potential and the challenges of implementing RSSI-based CDOA localization on a mobile robotic platform. The migration of the algorithm from ROS to ROS2, the deployment of multiple Raspberry Pi wireless nodes, and the integration of the robot-mounted access point have laid a strong foundation for later research experiments that are fully in ROS2 ecosystem. However, the results also demonstrate that wireless-based localization is highly sensitive to node synchronization and spatial layout. I feel like the number of nodes used has to scale depending on the environment's size.

The transition to a smaller mobile platform enables controlled testing conditions that are difficult to achieve on the larger rover. Issues such as IMU instability, unreliable wheel actuation, or inconsistent RSSI capture become easier to diagnose at this scale. These findings reinforce the need for a controlled development process: validating teleoperation reliability, verifying wireless-node communication, and ensuring the CDOA calculation pipeline is good under constrained, repeatable conditions. Overall, the work that I have completed to date provides essential insights into to the next steps required to achieve reliable wireless localization suitable for real-world poultry-farm environments.

VI. Future Work

Future work will focus on completing the small-scale experimental setup and progressively scaling the system toward deployment in a real poultry house. My immediate priority is to ensure that the mini robot used for testbed experiments is fully operational. Currently, the wheels are not responding to actuation commands, so resolving motor control, power distribution, and teleoperation functionality will be the first milestone. It kept on displaying "Waiting for signals from wheels...", but then crashed after 10-15 seconds. Once stable teleoperation is achieved, the wireless-node infrastructure will be integrated onto the smaller platform, allowing the full CDOA localization algorithm to be tested.

After validating the pipeline on the Husarion Rosbot, the system will be scaled to a larger indoor environment, such as our lab. This phase will evaluate how well the algorithm generalizes to increased distances, larger node spacing, and more complex motion trajectories. With robust performance demonstrated in the lab testing, the next step will involve developing autonomous navigation capabilities. CDOA-based localization will be fused with odometry or IMU data to enable closed-loop control, path following, and obstacle avoidance. This can be a great add-on to the robot because if the chicken farm is extremely large, maybe teleoperation is not a good idea.

Ultimately, the long-term objective is to deploy the system in an operational chicken farm. Once on-farm experiments begin, the robot will be used to investigate real-world challenges such as analyzing flock movement or scooping up dead chickens.

Acknowledgment

I would like to express my sincere gratitude to my Professor Ramviyas Nattanmai Parasuraman for his guidance and feedback throughout the development of this whole project. I would also like to thank my friend Daniel Laij for his support and collaboration during the debugging phases. Their contributions have been invaluable to my progress for this project.

VII. The use of Generative AI tool

I have used ChatGPT to help me polish some of my writing to make it sounds for professional. I did not use it to generate this report.

References

[1] E. Latif and R. Parasuraman, "Instantaneous Wireless Robotic Node Localization Using Collaborative Direction of Arrival," *IEEE Internet of Things Journal*, accepted for publication, 2023. Available: https://arxiv.org/abs/2307.01956.